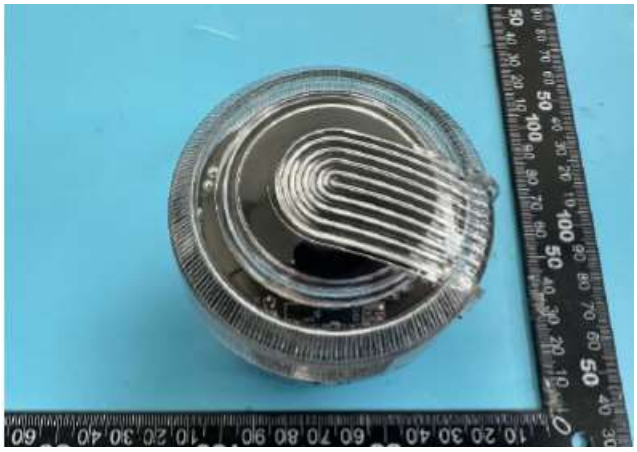
Prüfbericht-Nr.: *Test report no.*:	**CN236SSS 001**	**Auftrags-Nr.:** *Order no.*:	178186679	Seite 1 von 13 *Page 1 of 13*
Kunden-Referenz-Nr.: *Client reference no.*:	2518486	**Auftragsdatum:** *Order date:*	2023-08-25	
Auftraggeber: *Client*:	**Huizhou Duran Creative Technology Co., Ltd.** Room 708, Innovation Building, No. 1, Keji Road, Science and Technology Innovation Part, West Area of Daya Bay, Huizhou, Guangdong, P.R. China			
Prüfgegenstand: *Test item*:	Portable Light			
Bezeichnung / Typ-Nr.: *Identification / Type no.*:	INSTAFLASH V16 G-001			
Auftrags-Inhalt: *Order content*:	Issuing IP54 test report			
Prüfgrundlage: *Test specification*:	IEC 60529:1989+A1:1999+A2:2013 EN 60529:1991+A1:2000+A2:2013			

Wareneingangsdatum: *Date of sample receipt*:	2023-08-28
Prüfmuster-Nr.: *Test sample no:*	A003548920
Prüfzeitraum: *Testing period*:	2023-08-29 - 2023-08-29
Ort der Prüfung: *Place of testing*:	TUV Rheinland / CCIC (Qingdao) Co., Ltd.
Prüflaboratorium: *Testing laboratory:*	TUV Rheinland / CCIC (Qingdao) Co., Ltd.
Prüfergebnis*: *Test result**:	Pass



geprüft von: tested by:	X Geoff Guo Signed by: Geoff Guo	**genehmigt von:** *authorized by*:	X Signed by: LanHui Guo
Datum: *Date:* 2023-11-27		**Ausstellungsdatum:** *Issue date:* 2023-11-27	
Stellung / Position:	Geoff Guo / Project Engineer	**Stellung** / Position:	Guo Lanhui / Authorizer

Sonstiges / *Other*: The report is for IP54 test for the provided sample according to client's requirement.

Zustand des Prüfgegenstandes bei Anlieferung: *Condition of the test item at delivery*: Prüfmuster vollständig und unbeschädigt *Test item complete and undamaged*

* Legende:	P(ass) = entspricht o.g. Prüfgrundlage(n)	F(ail) = entspricht nicht o.g. Prüfgrundlage(n)	N/A = nicht anwendbar	N/T = nicht getestet
** Legend:*	*P(ass) = passed a.m. test specification(s)*	*F(ail) = failed a.m. test specification(s)*	*N/A = not applicable*	*N/T = not tested*

Dieser Prüfbericht bezieht sich nur auf das o.g. Prüfmuster und darf ohne Genehmigung der Prüfstelle nicht auszugsweise vervielfältigt werden. Dieser Bericht berechtigt nicht zur Verwendung eines Prüfzeichens.
This test report only relates to the a. m. test sample. Without permission of the test center this test report is not permitted to be duplicated in extracts. This test report does not entitle to carry any test mark.

Prüfbericht-Nr.: CN236SSS 001
Test report no.:

Anmerkungen
Remarks

1	Alle eingesetzten Prüfmittel waren zum angegebenen Prüfzeitraum gemäß eines festgelegten Kalibierungsprogramms unseres Prüfhauses kalibriert. Sie entsprechen den in den Prüfprogrammen hinterlegten Anforderungen. Die Rückverfolgbarkeit der eingesetzten Prüfmittel ist durch die Einhaltung der Regelungen unseres Managementsystems gegeben. Detaillierte Informationen bezüglich Prüfkonditionen, Prüfequipment und Messunsicherheiten sind im Prüflabor vorhanden und können auf Wunsch bereitgestellt werden. *The equipment used during the specified testing period was calibrated according to our test laboratory calibration program. The equipment fulfils the requirements included in the relevant standards. The traceability of the test equipment used is ensured by compliance with the regulations of our management system.* *Detailed information regarding test conditions, equipment and measurement uncertainty is available in the test laboratory and could be provided on request.*
2	Wie vertraglich vereinbart, wurde dieses Dokument nur digital unterzeichnet. Der TÜV Rheinland hat nicht überprüft, welche rechtlichen oder sonstigen diesbezüglichen Anforderungen für dieses Dokument gelten. Diese Überprüfung liegt in der Verantwortung des Benutzers dieses Dokuments. Auf Verlangen des Kunden kann der TÜV Rheinland die Gültigkeit der digitalen Signatur durch ein gesondertes Dokument bestätigen. Diese Anfrage ist an unseren Vertrieb zu richten. Eine Umweltgebühr für einen solchen zusätzlichen Service wird erhoben. Informationen zur Verifizierung der Authentizität unserer Dokumente erhalten Sie über folgenden Link: Einführung in digitale Signaturen *As contractually agreed, this document has been signed digitally only. TUV Rheinland has not verified and unable to verify which legal or other pertaining requirements are applicable for this document. Such verification is within the responsibility of the user of this document. Upon request by its client, TUV Rheinland can confirm the validity of the digital signature by a separate document. Such request shall be addressed to our Sales department. An environmental fee for such additional service will be charged. For information on verifying the authenticity of our documents, please visit the following link: Introduction to Digital Signature*
3	Prüfklausel mit der Note * wurden an qualifizierte Unterauftragnehmer vergeben und sind unter der jeweiligen Prüfklausel des Berichts beschrieben. Abweichungen von Prüfspezifikation(en) oder Kundenanforderungen sind in der jeweiligen Prüfklausel im Bericht aufgeführt. *Test clauses with remark of * are subcontracted to qualified subcontractors and descripted under the respective test clause in the report.* *Deviations of testing specification(s) or customer requirements are listed in specific test clause in the report.*
4	Die Entscheidungsregel für Konformitätserklärungen in diesem Prüfbericht basiert auf der "Null-Grenzwert-Regel" und der "Einfachen Akzeptanz" gemäß ILAC G8:2019 und IEC Guide 115:2021, es sei denn, in der auf Seite 1 dieses Berichts genannten angewandten Norm ist etwas anderes festgelegt oder vom Kunden gewünscht. Dies bedeutet, dass die Messunsicherheit nicht berücksichtigt wird und daher auch nicht im Prüfbericht angegeben wird. *The decision rule for statements of conformity in this test report is based on the "Zero Guard Band Rule" and "Simple Acceptance" in accordance with ILAC G8:2019 and IEC Guide 115:2021, unless otherwise specified in the applied standard mentioned on Page 1 of this report or requested by the customer. This means that measurement uncertainty is not taken in account and hence also not declared in the test report.*
5	Liste der Anhänge: *List of Attachment(s):* N/A

Produktbeschreibung
Product description

Summary of testing:

Tests performed (name of test and test clause):	**Test Location:**
Dust protection test IP5X: The EUT is suspended outside the dust chamber and operated at rated supply voltage until the operating temperature is achieved. Then the EUT, whilst still operating, is placed with the minimum disturbance in the dust chamber. The door of the dust chamber is closed. The blower causing the talcum powder to be in suspension is switched on. After 1 min, the luminaire is switched off and allowed to cool for 3 h whilst the talcum powder remains in suspension. Water protection test IPX4: The EUT was sprayed from every direction with water for 10 min by means of the oscillating tube. The EUT was mounted under the pivot line of the tube so that the ends of EUT receive adequate coverage from the jets. It was concluded that the EUT met the requirements of the degree of protection IP54. **Note:** EUT= Equipment Under Test	TUV Rheinland / CCIC (Qingdao) Co., Ltd. Floor 6, 4, 2&1, Building 2, No.175, Zhuzhou Road, Laoshan District, Qingdao, Shandong, China

Absatz *Clause*	**Anforderungen - Prüfungen /** *Requirements - Tests*	**Messergebnisse – Bemerkungen/** *Measuring results - Remarks*	**Ergebnis** *Result*

TEST REPORT
IEC 60529
Degrees of protection provided by enclosures (IP Code)

10	**Marking**		P
	Where appropriate, such a standard should also specify the method of marking which is to be used when – one part of an enclosure has a different degree of protection to that of another part of the same enclosure; – the mounting position has an influence on the degree of protection; – the maximum immersion depth and time are indicated.	IP 54 specified in EN 60598-2-4 & EN 60598-1	P

11	**General requirements for tests**		P
11.1	Atmospheric conditions for water or dust tests		P
	Unless otherwise specified in the relevant product standard, the tests should be carried out under the standard atmospheric conditions described in IEC 60068-1.		P
	The recommended atmospheric conditions during the tests are as follows:		P
	Temperature range: 15°C to 35°C	Considered	P
	Relative humidity: 25 % to 75 %	Considered	P
	Air pressure: 86 kPa to 106 kPa (860 mbar to 1060 mbar)	Considered	P
11.2	Test samples		P
	The tests specified in this standard are type tests		P
	Unless otherwise specified in a relevant product standard, the test samples for each test shall be in a clean and new condition, with all parts in place and mounted in the manner stated by the manufacturer.	New sample provided by the manufacturer.	P
	If it is impracticable to test the complete equipment, representative parts or smaller equipment having the same full-scale design details shall be tested.	Complete equipment tested	P
	The relevant product standard shall specify details such as:		P
	- the number of sample to be tested;	One sample	P
	- conditions for mounting, assembling and positioning of the samples, for example by the use of an artificial surface (ceiling, floor or wall)	Ceiling mounting	P
	- the pre-conditioning, if any, which is to be used;	As required in EN 60598-1, refer to Summary of testing on Page 3	P
	- whether to be tested energized or not;	As required in EN 60598-1, refer to Summary of testing on Page 3	P
	- whether to be tested with its parts in motion or not.		N/A

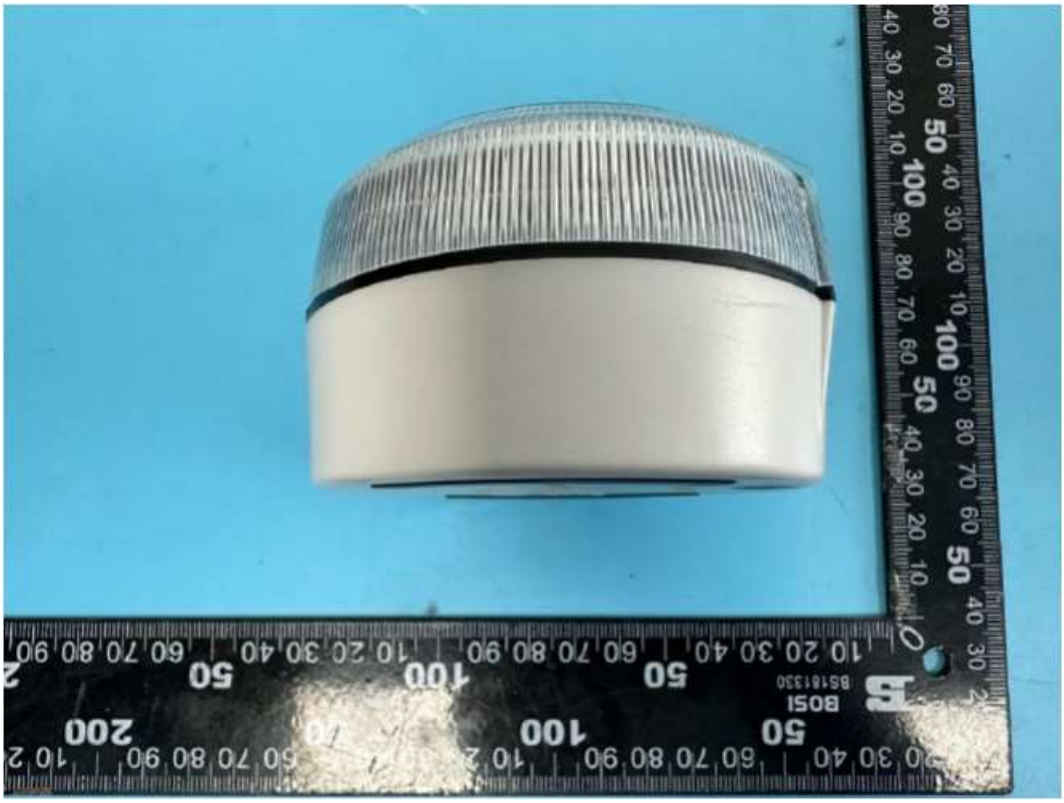
Absatz *Clause*	**Anforderungen - Prüfungen /** *Requirements - Tests*	**Messergebnisse – Bemerkungen/** *Measuring results - Remarks*	**Ergebnis** *Result*
	In the absence of such specification, the manufacturer's instructions shall apply.		P
12	**Tests for protection against access to hazardous parts indicated by the first characteristic numeral**		P
12.1	Access probes Access probes to test the protection of persons against access to hazardous parts are given in table 6		P
12.2	Test conditions The access probe is pushed against or (in case of the test for first characteristic numeral 2) inserted through any openings of the enclosure with the force specified in table 6. For tests on low-voltage equipment, a low-voltage supply (of not less than 40 V and not more than 50 V) in series with a suitable lamp should be connected between the probe and the hazardous parts inside the enclosure. Hazardous live parts covered only with varnish or paint, or protected by oxidation or by a similar process, are covered by a metal foil electrically connected to those parts which are normally live in operation. The signal-circuit method should also be applied to the hazardous moving parts of high-voltage equipment. Internal moving parts may be operated slowly, where this is possible.		P
12.3	Acceptance conditions		P
12.3.1	For low-voltage equipment (rated voltages not exceeding 1000 V a.c. and 1500 V d.c.) The access probe shall not touch hazardous live parts. If adequate clearance is verified by a signal circuit between the probe and hazardous parts, the lamp shall not light.	Test wire 1.0 mm diameter, 100 mm long cannot penetrate.	P
12.3.2	For high-voltage equipment (rated voltages exceeding 1000 V a.c. and 1500 V d.c.) When the access probe is placed in the most unfavourable position(s), the equipment shall be capable of withstanding the dielectric tests as specified in the relevant product standard applicable to the equipment. Verification may be made either by dielectric test or by inspection of the specified clearance dimension in air which would ensure that the tests would be satisfactory under the most unfavourable electric field configuration (see IEC 60071-2). In the case where an enclosure includes sections at different voltage levels the appropriate acceptance conditions for adequate clearance shall be applied for each section.		N/A

Absatz *Clause*	**Anforderungen - Prüfungen /** *Requirements - Tests*	**Messergebnisse – Bemerkungen/** *Measuring results - Remarks*	**Ergebnis** *Result*
12.3.3	For equipment with hazardous mechanical parts: The access probe shall not touch hazardous mechanical parts. If adequate clearance is verified by a signal circuit between the probe and hazardous parts, the lamp shall not light.	Not applicable	N/A
13	**Tests for protection against solid foreign objects indicated by the first characteristic numeral.**		P
13.1	Test means: Test means and the main test conditions are given in table 7.		P
13.2	Test conditions for first characteristic number 1, 2, 3, 4		N/A
13.3	Acceptance conditions for first characteristic number 1, 2, 3, 4		N/A
13.4	Dust test for first characteristic number 5 and 6		P
	Enclosures are of necessity in one of two categories: - Category 1: Enclosures where the normal working cycles of the equipment causes reductions in air pressure within the enclosure below that of the surrounding air, for example, due to thermal cycling effects. - Category 2: Enclosures where no pressure difference relative to the surrounding air is present.	Category 2	P
	Category 1 enclosures:		N/A
	Category 2 enclosures:		P
	Category 1 and Category 2 enclosures:	Complete product tested	N/A
13.5	Special conditions for first characteristic number 5		P
13.5.1	Test conditions for first characteristic number 5		P
13.5.2	Acceptance conditions for first characteristic number 5		P
13.6	Special conditions for first characteristic number 6		N/A
13.6.1	Test conditions for first characteristic number 6		N/A
13.6.2	Acceptance conditions for first characteristic number 6		N/A
14	**Tests for protection against water indicated by the second characteristic numeral**		P
14.1	Test means Test means and the main test conditions are given in table 8.		P
14.2	Test conditions		P
14.2.1	Test for second characteristic numeral 1 with the drip box		N/A
14.2.2	Test for second characteristic numeral 2 with the drip box		N/A
14.2.3	Test for second characteristic numeral 3 with oscillating tube or spray nozzle		N/A

Absatz *Clause*	**Anforderungen - Prüfungen /** *Requirements - Tests*	**Messergebnisse – Bemerkungen/** *Measuring results - Remarks*	**Ergebnis** *Result*
14.2.4	Test for second characteristic numeral 4 with oscillating tube or spray nozzle	IPX4 applied	P
14.2.5	Test for second characteristic numeral 5 with the 6.3 mm nozzle		N/A
14.2.6	Test for second characteristic numeral 6 with the 12.5 mm nozzle		N/A
14.2.7	Test for second characteristic numeral 7: temporary immersion between 0,15 m and 1 m		N/A
14.2.8	Test for second characteristic numeral 8: continuous immersion subject to agreement		N/A
14.2.9	Test for second characteristic numeral 9 by high pressure and temperature water jetting		N/A
14.3	Acceptance conditions After testing in accordance with the appropriate requirements of 14.2.1 to 14.2.9, the enclosure shall be inspected for ingress of water.		P
15	**Tests for protection against access to hazardous parts indicated by the additional letter**	No additional letter applied	N/A
15.1	Access probes Access probes to verify the protection of persons against access to hazardous parts are given in table 6.		N/A
15.2	Test conditions The access probe is pushed against any openings of the enclosure with the force specified in table 6. If it partly or fully penetrates, it is placed in every possible position, but in no case shall the stop face fully penetrate through the opening. Internal barriers are considered part of the enclosure as defined in 3.1.		N/A
15.3	Acceptance conditions The protection is satisfactory if adequate clearance is kept between the access probe and hazardous parts.		N/A

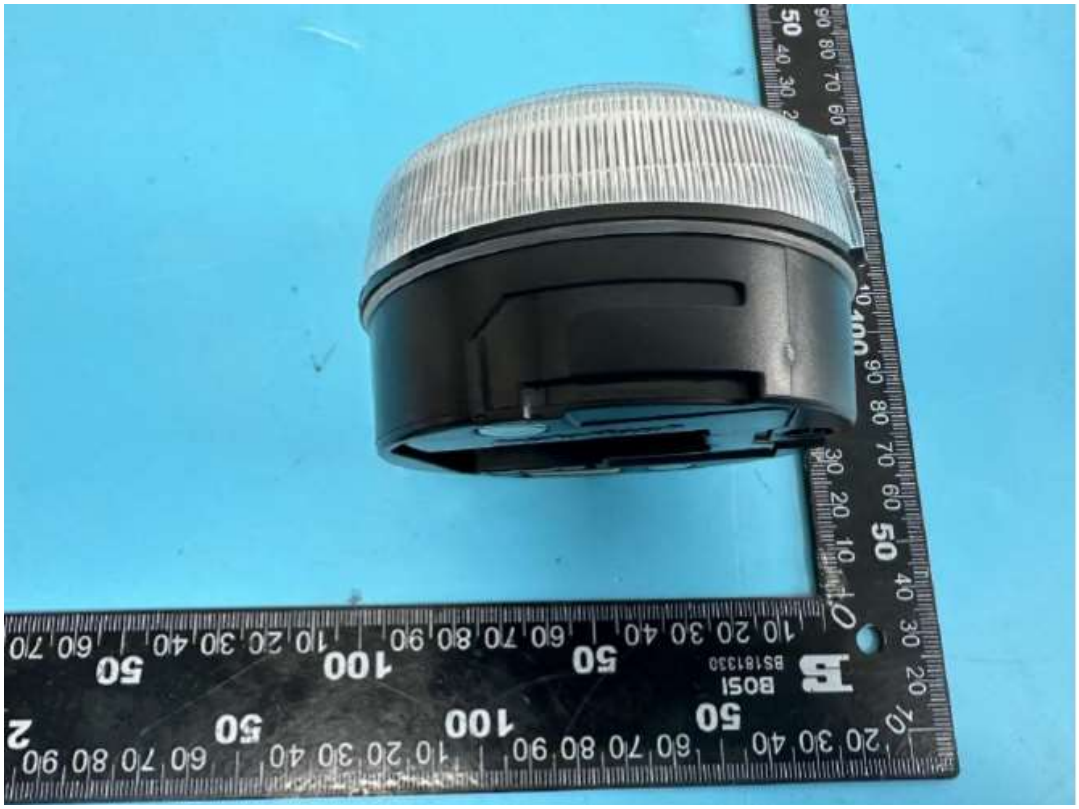
Absatz *Clause*	**Anforderungen - Prüfungen /** *Requirements - Tests*	**Messergebnisse – Bemerkungen/** *Measuring results - Remarks*	**Ergebnis** *Result*





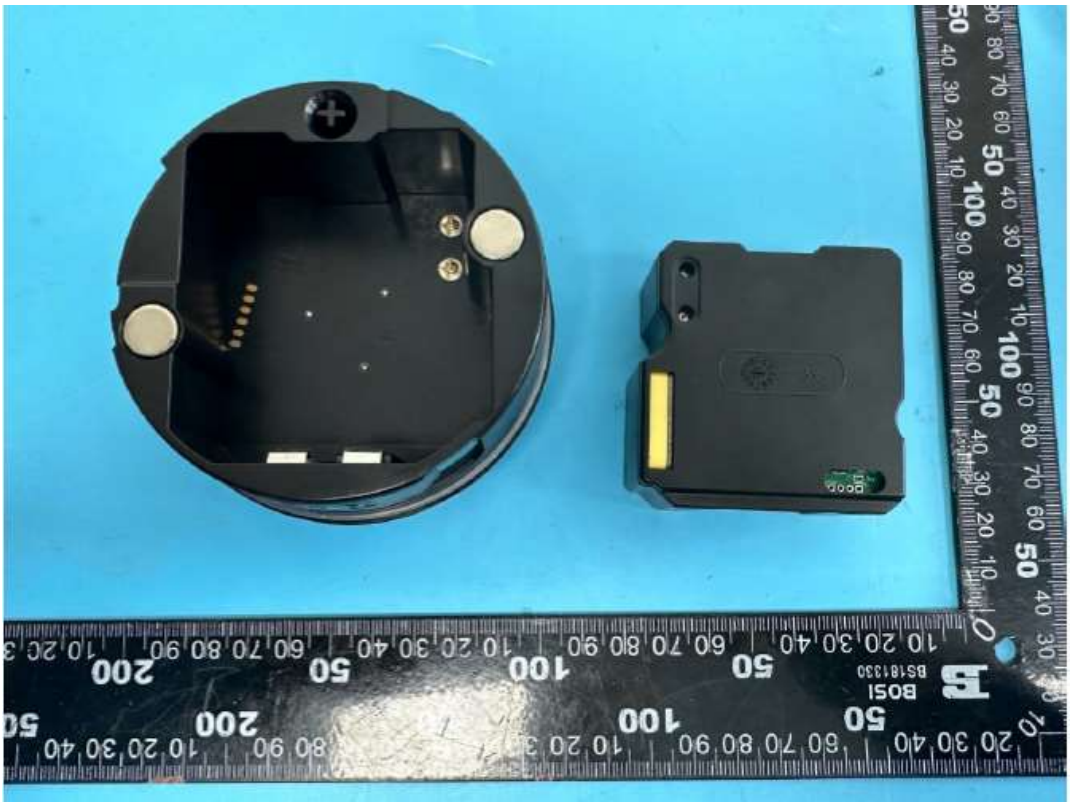
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Absatz *Clause*	**Anforderungen - Prüfungen /** *Requirements - Tests*	**Messergebnisse – Bemerkungen/** *Measuring results - Remarks*	**Ergebnis** *Result*





Absatz *Clause*	**Anforderungen - Prüfungen /** *Requirements - Tests*	**Messergebnisse – Bemerkungen/** *Measuring results - Remarks*	**Ergebnis** *Result*

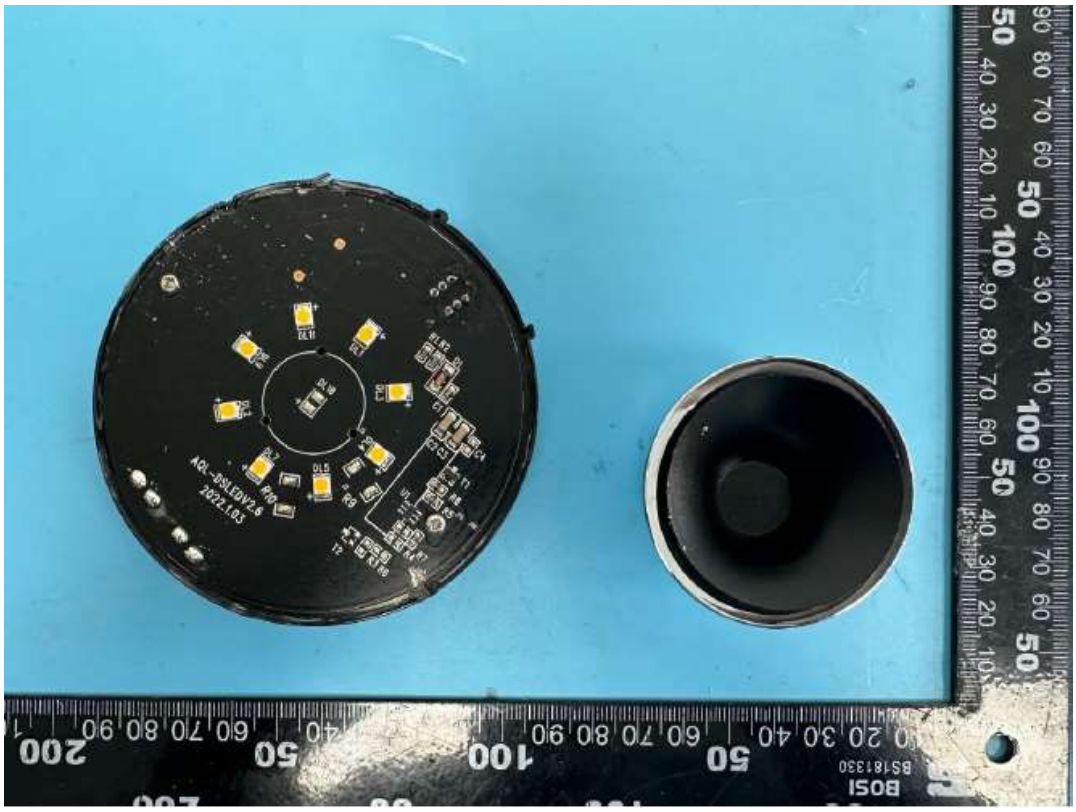




Absatz *Clause*	**Anforderungen - Prüfungen /** *Requirements - Tests*	**Messergebnisse – Bemerkungen/** *Measuring results - Remarks*	**Ergebnis** *Result*



Cystal



Absatz *Clause*	**Anforderungen - Prüfungen /** *Requirements - Tests*	**Messergebnisse – Bemerkungen/** *Measuring results - Remarks*	**Ergebnis** *Result*



